PixelStacks: High-Capacity Image Steganography for Censorship Circumvention of Long-Form Content

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ABSTRACT

This work presents a novel approach to circumvent internet censorship by leveraging image steganography to transmit long-form sensitive content, such as books, essays, and news articles, across censored networks. At the core of this work is an advanced steganography algorithm that achieves high data capacity while maintaining imperceptible noise levels in stego-images. The algorithm combines two key innovations: pseudorandom channel-based least significant bit (LSB) embedding and adaptive matching to reduce pixel distortion. Through evaluations using peak signal-to-noise ratio (PSNR) and steganalysis metrics, our method demonstrates superior performance compared to existing approaches. To showcase its practical application, we developed a web-based library tool that enables users to access censored long-form content while preserving the privacy of the transmitted data. Unlike well-established censorship circumvention methods such as VPNs, TOR, or encryption, that are subjected to heavier counter methods by censorship authorities, this tool leverages steganography to conceal the nature of the transmitted content, allowing users to bypass packet inspection-based censorship mechanisms. By focusing on data-level privacy and accessibility, this work offers a novel framework for privacy-enhancing technology designed for real-world applications in restrictive environments.

KEYWORDS

Censorship, Steganography

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1 INTRODUCTION

The suppression of information remains a global challenge, with authorities frequently restricting access to politically or socially sensitive materials [1, 2]. While digital platforms have provided alternative channels to circumvent physical censorship [3], sophisticated keyword-based content filtering systems have rendered even online dissemination vulnerable to interference. For individuals in heavily censored regions, developing effective methods to bypass these restrictions is essential for ensuring access to and the dissemination of restricted content.

Traditional approaches to combating keyword-based censorship, such as substitution techniques, involve replacing censored terms with inline images or homophones [4, 5]. While these methods are effective for short-form content like blog posts or comments, they impose a significant cognitive burden on readers when applied to long-form content, such as books or news articles. Encryption-based solutions, such as VPNs and TOR, offer robust alternatives by obfuscating network traffic. However, these methods are not universally viable. In some regions, VPNs have been outlawed [6], and TOR traffic, identifiable through deep packet inspection (DPI), is actively filtered by network censors [7]. Moreover, encrypted protocols such as Shadowsocks and VMess, which aim to appear indistinguishable from random traffic, have also been blocked en masse based on passive traffic analysis, as seen in recent advancements within China's Great Firewall (GFW) [8]. These counter-circumvention studies underscore the need for developing alternative censorship circumvention techniques that goes outside of the range of traditional methods.

To address these challenges, researchers have increasingly explored steganography as a viable method for covertly transmitting sensitive information over censored networks [9–12]. Steganography enables the concealment of sensitive data within seemingly innocuous cover media, such as images, without alerting network censors. By encoding sensitive text in a way that evades detection during transmission and allowing seamless decoding by recipients, steganography provides an alternative to encryption, VPNs, and TOR, while mitigating the accessibility barriers associated with those methods.

Building on this body of work, we propose leveraging high-capacity steganography to facilitate the distribution of long-form sensitive content, such as books and articles, across censored networks. High payload capacity is critical for transmitting extensive textual content, particularly for non-English languages where characters often require more than one byte for encoding. Moreover, the censorship of prominent social media platforms in some countries poses additional challenges [13–15], as users cannot rely on proxy systems dependent on restricted platforms [16]. To address these issues, we aim to design a lightweight, web-based platform capable of encoding and decoding steganographic content, creating a resilient backbone for an online steganographic library.

The platform's design must account for the mechanisms underlying keyword censorship, such as TCP packet injection [17], wherein censors disrupt connections by injecting or terminating packets in the TCP stream [18]. To avoid these surveillance-triggered methods, our tool enables users to perform encoding and decoding operations locally, reducing reliance on server-side processing. This approach presents unique challenges, as modern steganography algorithms, particularly those employing machine learning techniques [19–22], often demand computational resources that exceed the capabilities of mobile devices. Given that mobile phones are increasingly the primary computing devices for much of the global population [23], designing lightweight algorithms capable of operating efficiently on resource-constrained devices is imperative. Additionally, current

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steganography implementations often lack support for non-English character encodings, a limitation that undermines their applicability in diverse linguistic contexts [24]. To ensure inclusivity, our platform supports broad Unicode character sets, addressing the global nature of censorship.

With these considerations, this paper makes the following contributions:

(1) We present the design and implementation of a novel steganography algorithm capable of carrying large amounts of text data while minimizing pixel distortion using pseudorandom channel-based Least Significant Bit (LSB) embedding and adaptive matching. The algorithm, along with practical features such as Unicode support and client-side processing, is publicly available.
(2) We evaluate the performance and security of our proposed PRNG-based method against analogous steganography techniques using metrics like peak signal-to-noise ratio (PSNR) and sample pair analysis.
(3) We validate the viability of steganography for transmitting sensitive, long-form text content through experiments conducted on WeChat, a well-known censored platform.

The remainder of this paper is organized as follows. Section Background and Related Work reviews related work on online censorship mechanisms, censorship circumvention, and steganography-based approaches. Section Methodology outlines the design goals and implementation of our steganography algorithm and web prototype. Section Experiment presents the evaluation of our method's performance and its application in transmitting sensitive content across censored networks. Section Discussion discusses the experimental findings and the limitations of this research and identifies potential directions for future work.

2 BACKGROUND AND RELATED WORK

2.1 Background

2.1.1 Online Censorship Mechanisms. Extensive research has been conducted to reverse-engineer the mechanisms underlying information censorship in various nation-states. Sun et al. demonstrated how the Chinese government delegates content screening responsibilities to private internet companies [25]. The Citizen Lab has extensively analyzed platform-specific censorship, revealing techniques such as hash indexing of shared images on WeChat for real-time filtering [26] and selective keyword-based censorship on chats involving mainland China-registered accounts [27]. Additionally, studies have shown that different search engines enforce censorship practices inconsistently across China [28]. These efforts provide critical technical insights into censorship infrastructure, laying the groundwork for the development of effective circumvention techniques. Similar mechanisms, such as TCP packet injection for traffic disruption, have also been observed in other regions [17, 18].

2.1.2 Basics of Steganography. Steganography conceals the existence of a message by embedding it within a "cover object" (e.g., an image, audio file, or text) before transmission. This contrasts with cryptography, which transforms a message into an unintelligible form. Spatial domain techniques, such as Least Significant Bit (LSB) insertion, embed binary data directly into pixel values, providing high capacity and low computational complexity [29]. However, these methods are susceptible to detection via steganalysis techniques like RS analysis and visual attacks [30]. Enhanced techniques, such as LSB matching with randomized ±1 modification, address these vulnerabilities by mitigating repetitive patterns in pixel value changes [31]. Recent advances have further refined steganography, employing methods like edge detection for embedding area selection and channel-dependent payload partitioning [32, 33].

2.2 Related Work

2.2.1 Censorship Circumvention. Efforts to bypass keyword-based censorship have led to the development of substitution techniques that replace sensitive words with alternatives comprehensible to human readers but obscure to automated filters. Patil introduced a mechanism that substitutes banned keywords with rare alternatives accompanied by a decoding dictionary [5], while Hiruncharoenvate et al. leveraged homophones to replace censored characters [4]. These methods, while effective for short-form content, impose cognitive burdens on users and are ill-suited for long-form content like books or essays.

Proxy-based services, such as Psiphon, Freegate, and Ultrasurf, use dynamic IP rotation to evade censorship [34–36]. However, these services face limitations, as their names and functionalities are often blocked by keyword filters [37]. Furthermore, VPNs and TOR, while robust, are restricted in some regions [6] and easily identifiable through DPI [7], limiting accessibility for less technically literate users.

2.2.2 Steganography as Censorship Circumvention. Steganography offers a promising alternative for censorship circumvention, as it preserves the legibility of the transmitted data. Invernizzi et al. introduced "Message in a Bottle," a protocol for embedding messages in arbitrary online locations to establish covert communication channels [10]. Connolly et al. developed TRIST, a system that embeds data in the frequency domain of images, ensuring resilience to transformations like transcoding [9]. Similarly, Barradas et al. proposed Stegozoa, which embeds covert data into WebRTC video streams to prevent detection during gateway inspections [11]. These works demonstrate the adaptability of steganography for censorship resistance but often lack the carrying capacity necessary for long-form text.

2.2.3 PRNG-based Steganography. Pseudorandom Number Generator (PRNG)-based approaches enhance the security and robustness of steganography by dispersing payloads across cover media in a randomized manner. Alam et al. employed multiple PRNGs to determine the color channels, payload bit counts, and pixel order for embedding [38], while Ehsan et al. used PRNGs to randomize embedding locations and XOR-based obfuscation for payload bits [39]. These techniques strike a balance between computational efficiency and resilience against detection, making them ideal candidates for censorship circumvention.

3 METHODOLOGY

We designed an image steganography algorithm to address three critical requirements:

(1) Enable a cover image to carry a large payload to support long-form content transmission;
(2) Minimize pixel distortion to enhance security and reduce detectability by steganalysis tools;
(3) Ensure embedding and extraction processes are computationally efficient for feasible client-side processing.

The proposed steganographic algorithm focuses on enhancing embedding capacity and security while minimizing distortion in the cover image. Traditional LSB insertion methods often concentrate pixel distortions in localized regions, making stego-images more susceptible to detection by modern steganalysis techniques. To overcome these limitations, we introduce two novel techniques: a pseudorandom channel-based embedding approach to distribute payload across color channels more evenly and an adaptive matching technique to minimize pixel value distortions further.

To ensure practicality, we chose to forego the latest machine learning-based steganographic techniques, which are computationally intensive and unsuitable for client-side processing on resource-constrained devices. Our approach prioritizes practicality and accessibility while ensuring robust security features.

Additionally, we implemented a prototype web application that integrates the proposed steganographic techniques. This proof-of-concept system demonstrates the feasibility and practicality of our approach for censorship circumvention.

3.1 Pseudorandom Channel-Based Embedding

The proposed method generates a pseudorandom sequence to select values for embedding the payload. To maximize carrying capacity, it employs a 3-3-3 bit allocation strategy, distributing one byte of payload across the three color channels (R, G, and B) of different pixels. Unlike the conventional 3-3-2 strategy, which sacrifices one bit of storage for ease of implementation, the 3-3-3 strategy utilizes all available bits, increasing capacity. While this approach introduces additional complexity, it is well-suited for high-capacity steganographic systems. Furthermore, distortion per pixel is mitigated using the Adaptive Matching technique, as detailed in the next subsection, ensuring security and imperceptibility.

Rather than using the pseudorandom sequence to directly select entire pixels for embedding, our approach treats individual color channels as the fundamental units. For example, previous techniques leveraging pseudorandom pixel selection schemes [40, 41] would, for a 512×512 RGB image with 262, 144 pixels, use the pseudorandom sequence to select one pixel and embedding one byte of data across all three color channels of that pixel. This method, however, concentrates distortion within individual pixels. By contrast, our method considers each of the 3 color channels in a pixel as a potential embedding unit, dispersing payload bits across the color channels of different pixels. As a result, one byte of data is embedded in the R, G, and/or B channels of likely separate pixels, diluting distortion and enhancing randomness in the payload distribution throughout the cover image. This approach is particularly beneficial when embedding capacity is not fully utilized, as it creates a more balanced distribution of altered and unaltered pixel values.

We employ the Fisher-Yates Shuffle algorithm to generate the pseudorandom number sequence for channel selection, using a shared random seed key for predictable and repeatable sequences during the embedding and extraction processes. The Fisher-Yates Shuffle is recognized for producing unbiased permutations efficiently [42], a critical feature for ensuring the security and repeatability of hidden data retrieval. Algorithm 1 presents the pseudocode implementation of the Fisher-Yates Shuffle algorithm:

Algorithm 1 Fisher-Yates Shuffle

Input: $A_1 \dots A_N$
Output: Shuffled Array A

```
for i ← N to 1 do
    k ← random integer such that 0 <= k <= i
    temp ← A[i]
    A[i] ← A[k]
    A[k] ← temp
end for
```

3.2 Adaptive Matching for Minimized Pixel Distortion

LSB Matching refers to the technique of randomly adding or subtracting from the value of the cover pixel if the LSB does not match the value of the payload bit [43]. For example, a bit string '1000' can be altered to carry the payload bit '1' by adding one so that it becomes '1001,' or by subtracting one to obtain '0111.' This simple alteration balances embedding distortion in the stego image, making the payload harder to detect through steganalysis.

Building on this foundation, we propose an adaptive matching technique that minimizes value imbalances in pixels by adjusting the embedding process. Utilizing 3 LSBs per color channel for payload embedding can introduce significant distortion to color values, making the stego image less robust against steganalysis. Adaptive Matching mitigates this issue by dynamically reducing the maximum distortion introduced to a color channel.

The process starts with a value from the original color channel, denoted as v_{ori}. When payload bits are embedded, this value changes to v_{emb}. The difference between these two values, δ, is calculated as:

$$\delta = v_{emb} - v_{ori}$$

If the magnitude of δ exceeds 2^{N-1} (where N is the number of LSBs used for embedding), we modify its sign to reduce its magnitude. This adjustment ensures that the distortion introduced by embedding remains minimal. In plain terms:

(1) If the difference δ is too large (greater than a threshold), the algorithm flips its sign to reduce its magnitude.
(2) For boundary cases, where δ equals the threshold, the algorithm randomly selects an adjustment value to maintain balance.
(3) Otherwise, no additional changes are made to δ.

Formally, we define a transformation function $f(\delta_{bin})$ as follows:

$$f(\delta_{bin}) \begin{cases} -(a_{N-1}-1)^2 2^{N-1} + \sum_{i=1}^{N-2} a_i 2^i, & \text{if } abs(\delta) > 2^{N-1} \quad (1) \\ \text{select (1) or (3) randomly,} & \text{if } abs(\delta) = 2^N \quad (2) \\ -a_{N-1}2^{N-1} + \sum_{i=1}^{N-2} a_i 2^i, & \text{otherwise} \quad (3) \end{cases}$$

Adding $f(\delta_{bin})$ to the original channel value v_{ori} results in an adjusted value, v_{adp}, calculated as:

$$v_{adp} = v_{ori} + f(\delta_{bin})$$

Table 1 illustrates examples of this process, showing how Adaptive Matching reduces maximum pixel distortion. For instance, a normal 3-LSB insertion can introduce distortions of up to ±7, whereas our method reduces this to ±4, a 40% reduction in the maximum impact.

Algorithm 2 Adaptive Matching for 3-LSB payload

Input: Data-embedded binary pixel value v_{emd}, original binary pixel value v_{ori}
Output: Adapted pixel value v_{adp}

```
Δ ← v_emd − v_ori
if abs(Δ) > 4 then
    sign ← −1 if delta > 4 else 1
    match ← 8 mod abs(Δ)
    v_adp ← v_emd + (sign × match)
else if abs(Δ) = 4 then
    match ← Randomly select from +4 or −4
    v_adp ← v_emd + match
else
    v_adp ← v_emd
end if
```

3.3 Proof-of-concept Web Application

We have designed and developed an online platform that integrates our proposed steganographic techniques with additional functionalities, such as AES encryption and universal language support. The platform's primary design goal is to provide censored content to users in a way that is seamless and user-friendly, hiding the technical complexity of the decoding processes. Figure 3 illustrates the data flow for both the sender and receiver workflows. The web application can be accessed at https://bit.ly/hide-text.

3.3.1 User Interaction and Decoding Workflow. The platform features a gallery-like homepage displaying images that represent different books or other censored materials. When a user selects an image from the gallery, the decoding process is triggered automatically in the background, retrieving the embedded content associated with the image. The workflow proceeds as follows:

(1) Image Selection: The user clicks on an image in the gallery. This action initiates the decoding process, requiring no additional user input or technical knowledge.
(2) Seed Extraction: The embedded seed is extracted using a secondary pseudorandom sequence derived from a predefined key.
(3) Payload Extraction: The pseudorandom sequence guides the retrieval of payload bits from the image.
(4) Content Display: Once the extraction process completes, the decoded content (e.g., text from censored literature) is displayed to the user directly within the browser.

This automated decoding process ensures a seamless user experience, making the platform accessible even to non-technical users. The technical complexities of decoding and decryption remain hidden, while the tool transparently delivers censored content.

3.3.2 Encoding Workflow (Pre-Processing). The encoding workflow is handled during the preparation phase, where the cover images are embedded with payloads before being uploaded to the gallery. This process involves the following steps:

(1) Content Preparation: A message or book content is selected and encrypted with AES for security.
(2) Pseudorandom Sequence Generation: A pseudorandom number sequence is generated using the Fisher-Yates shuffle algorithm, with a unique seed to determine the order of color channel selection.
(3) Seed Embedding: The pseudorandom sequence's seed is embedded into the image using a secondary pseudorandom sequence.
(4) Payload Embedding: The pseudorandom sequence identifies specific color channels for embedding, with adaptive matching applied to minimize pixel distortion and preserve image quality.

By separating the encoding and decoding processes, the platform ensures that the decoding process remains invisible to the user while the encoding is handled securely during pre-processing.

We provide a step-by-step instruction for embedding a message into a cover image for reproducing the procedure of the web platform:

(1) The sender provides a mesaage, a cover image, and a key.
(2) Using the key, the message gets encrypted using AES.
(3) A pseudorandom number sequence of the length of $N \times N \times 3$ is generated the Fisher-Yates shuffle for an image of size $N \times N$. We utilize a time-dependantm unique seed for every request so that the pseudorandom number sequence does not repeat for images with the same size.
(4) The seed is then embedded in the image using a second pseudorandom number sequence using the key as the seed.
(5) Using the first pseudorandom number sequence, we select the corresponding color channel.
(6) Obtain the channel value if we were to embedd the payload bits as in a normal LSB insertion.
(7) Use the channel value with payload embedded to determine if adaptive matching could be applied, and if so, assign the adapted channel value to it to minimize pixel distortion.
(8) Repeat (5) - (7) until all payload bits are embedded.

The sender, upon obtaining the stego-image, can now distribute the image digitally. The distribution could take place in personal chats, or through an image hosting platform where the hidden content of an image is extracted locally and automatically in the receiver's browser. The instruction to extract hidden message from a stego-image is as follows:

Table 1: Examples of Adaptive Matching for Minimized Pixel Distortion

Original Value	Payload	Embedded Value	$(\delta)_2$	$(\delta)_{10}$	Adapted Value	$(\delta_{adp})_2$	$(\delta_{adp})_{10}$
01000000	111	01000111	0111	+7	00111111	1111	−1
01000000	110	01000110	0110	+6	00111110	1110	−2
01000000	101	01000101	0101	+5	00111101	1101	−3
01000000	100	01000100	0100	+4	00111100	1100	−4
01000111	011	01000011	1100	−4	01001011	0100	+4
01000111	000	01000000	1001	−7	00111000	0001	+1
01000111	001	01000001	1010	−6	00111001	0010	+2
01000111	010	01000010	1011	−5	00111010	0011	+3

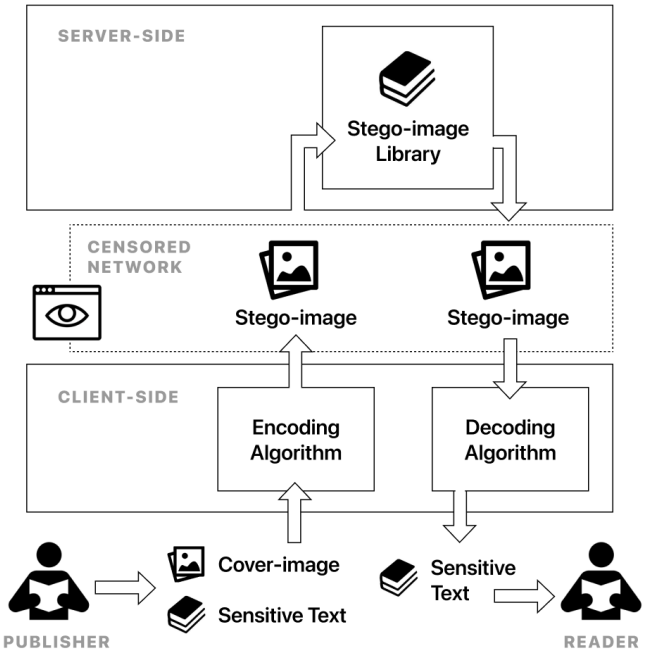

Figure 1: Endcoding and decoding flow in the web tool. (https://bit.ly/hide-text).

(1) The receiver provides a stego-image and a key.
(2) The embedded seed is extracted using a pseudorandom number sequence generated with Fisher Yates Shuffle using the key as the seed.
(3) Using the extracted seed, obtain the corresponding color value and read the LSBs.
(4) Collect LSB3 from pixel value until a unique stop sequence is detected.

Another design consideration that guided the web application architecture is that following a conventional client-server architecture exposes the sensitive content to the scrutiny of the network censors. When a web client interacts with an online application, the client typically transmit data over the network to be processed on the server-side. This client-server communication model offers several key benefits. By offloading computationally intensive tasks to the server infrastructure, it reduces the resource requirements on the client devices, making it well-suited for mobile platforms or legacy systems with limited processing capabilities. However, this architecture also introduces potential security and privacy concerns. By routing sensitive user information through the network, the data becomes exposed to any intermediaries, such as censors or eavesdroppers, present along the communication path. This exposure may render the client-server approach unsuitable for web applications servicing users located in heavily censored regions, where preserving user privacy is of paramount importance.

To address the security and privacy concerns inherent in the client-server architecture, we have opted for a client-side processing approach. By performing the encoding and decoding on the client device rather than routing sensitive information through the network, we can mitigate the risks of data exposure to potential censors or eavesdroppers. While acknowledging the advancements enabled by machine learning in the field of steganography, it would be difficult to deploy those methods in censorship circumvention where traditional client-server communication might expose sensitive information to censors, and users might not have adequate computers to execute intensive processes. Additionally, the web tool includes AES encryption, allowing users to set a key to encrypt the payload before embedding it into the cover media. This added encryption layer further enhances the security and privacy of the sensitive information being transmitted.

In addition to the security and privacy considerations, our web tool also supports universal language encoding and decoding. By leveraging the UTF-8 character encoding standard, the tool can handle text in a wide range of languages, including non-Latin scripts such as Chinese, Cyrillic, and Arabic. This universal language support ensures that the steganographic capabilities provided by our method are accessible to a global audience.

This brings us to a central focus of this research - to develop a practical steganography tool that can assist users in circumventing book censorship. By implementing a client-side processing approach and a lightweight yet high carrying capacity steganographic algorithm, and offering encryption capability, we have created a solution that balances robust security guarantees with accessible usability, even on resource-constrained client devices. Crucially, the web-based tool supports universal language encoding, allowing users across diverse linguistic backgrounds to leverage its steganographic capabilities. This design choice reflects a deliberate trade-off between technical sophistication and real-world practicality, ensuring the tool remains a viable option for those located

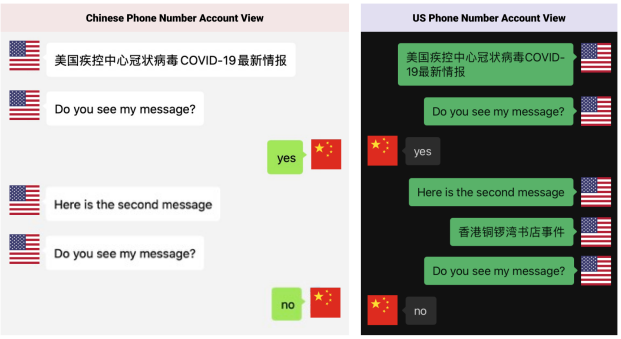

Figure 2: Two screenshots of WeChat with profile picture covered and device identifying information removed.

in regions with heavy internet censorship, where the ability to covertly transmit information is most needed.

4 EXPERIMENT

This section evaluates the proposed method across two key aspects. First, we test the effectiveness of using steganographic images to transmit lengthy text content in bypassing keyword-based content filtering. Second, we assess the robustness of the proposed method against steganalysis techniques, focusing on how our innovations—pseudorandom channel-based embedding and adaptive matching—improve security and image quality compared to existing methods.

4.1 Censorship Evasion on WeChat via Steganography

The goal of this experiment is to validate the hypothesis that our steganographic method can effectively transmit long-form text containing censored keywords over networks with keyword-based censorship. We selected WeChat as the testing platform due to its well-documented censorship mechanisms, including deep packet inspection (DPI), and its prominence as a communication tool in heavily censored regions. Previous research conducted by Citizen Lab has provided detailed insights into WeChat's server-side censorship rules, which block messages containing blacklisted keywords before they reach the recipient [27, 28, 44].

4.1.1 Keyword Identification. To ensure relevance, we first identified keywords actively censored on the WeChat network at the time of testing. Previous studies have shown that the keyword lists maintained by platforms like WeChat are updated periodically, often reflecting current political and social events [37]. Starting with a list of known censored keywords maintained by Citizen Lab (https://github.com/citizenlab/chat-censorship), we systematically tested each keyword by sending plain text messages from a US-registered WeChat account to a China-registered account. Messages containing uncensored keywords were delivered successfully, while censored keywords were blocked. Figure 2 shows an example where a message containing the keyword "Hong Kong Causeway Bay Bookstore Incident" was blocked, while a message containing "US Centers for Disease Control and Prevention COVID-19 latest news" was successfully delivered.

4.1.2 Transmission of Steganographic Images. In the second phase of the experiment, we evaluated the ability of steganographic images to bypass keyword-based censorship on WeChat. Using the web tool, we embedded five censored keywords into images as payloads. These images were then sent from the US-registered WeChat account to the China-registered account. All steganographic images were received successfully, and the embedded payloads were extracted without triggering any keyword-based filtering.

To further test the method's carrying capacity, we created another set of steganographic images containing entire articles discussing events associated with the censored keywords. Each article was repeated to maximize the image's carrying capacity. These images were also successfully transmitted and decoded, confirming that WeChat did not apply compression or transformations to images sent with the "Full Image" option, ensuring the integrity of the embedded payloads.

Figure 3: Two screenshots of WeChat (with profile picture covered and device identifying information removed) showing the successful decoding of the image sent through WeChat from a client at a vantage point.

This experiment demonstrates that the proposed steganographic method can effectively bypass keyword-based censorship on a platform like WeChat, even when transmitting large payloads.

4.2 Steganalysis

We conducted a comprehensive evaluation of the proposed method's robustness against steganalysis techniques. The experiment compared three approaches:

(1) Baseline: Sequential 3-LSB insertion.
(2) Intermediate: 3-LSB insertion with random pixel selection using the Fisher-Yates shuffle.
(3) Proposed Method: Intermediate approach with pseudorandom channel-based embedding and adaptive matching.

4.2.1 Evaluation Metrics and Dataset. We used the RGB-BMP Steganalysis Dataset created by Mudhafar [45]. There are a total of 3000 bitmap images in this dataset, each iamge has the dimension of 512 times 512. We embedded text data into these images using three methods, (1) sequential LSB insertion, (2) PRNG-enabled pixel selection and insertion, and (3) our proposed method. To examine how these different embedding algorithms perform with different amount of hidden data, we also embed data at 25%, 50%, 75%, and 100% capacity for each of the embedding methods.

Steganalysis metrics were calculated using OpenCV and StegExpose [46], including Sample Pairs [47], RS Analysis [30], and Fusion [48], Chi Square Attack [49], and Peak signal-to-noise ratio.

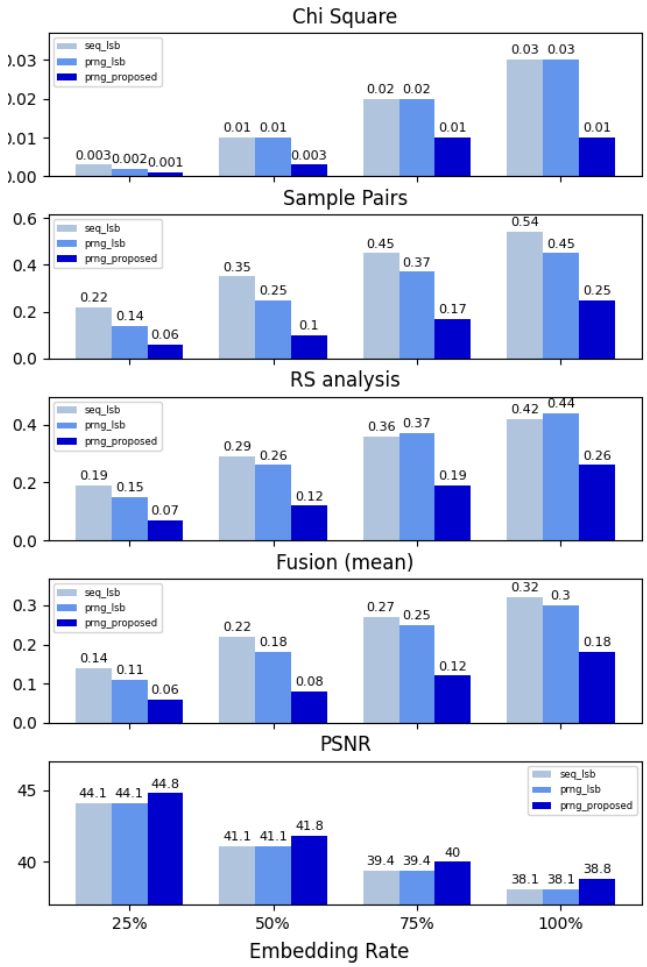

Figure 4: Performance comparison of different steganographic methods (seq lsb, prng lsb, and prng proposed) across various embedding rates. The evaluation metrics include Chi-Square, Sample Pairs, RS Analysis, Fusion (mean), and PSNR, demonstrating the impact of increasing payload on statistical detectability and image quality.

4.2.2 Results and Observations. The Chi-Square analysis serves as a statistical measure to detect anomalies in the distribution of pixel values introduced by data embedding. A lower Chi-Square value indicates a steganographic method's effectiveness in avoiding statistical irregularities, thereby reducing detectability. The proposed method consistently achieves the lowest Chi-Square values across all embedding rates. Its advantage over the sequential method and the PRNG-only method is amplified as the embedding rate increases. Specifically, it records a value of 0.001 at a 25% embedding rate and maintains a reduced value of 0.01 for embedding rates of 50% and higher. These results demonstrates the proposed method's capacity to obscure statistical artifacts, enhancing its robustness against conventional Chi-Square attacks.

The Sample Pairs analysis evaluates changes in the correlation between adjacent pixel pairs, which is a key indicator of the detectability of hidden data. A steganographic method that minimally disrupts pixel correlations is less likely to be detected. The proposed method demonstrates superior performance by maintaining the lowest Sample Pairs values across all embedding rates. At a 25% embedding rate, it achieves a value of 0.06, significantly lower than the 0.22 recorded for sequential embedding method and 0.14 for the PRNG only, intermediate method. Even at a 100% embedding rate, the proposed method sustains a lower value of 0.25, in stark contrast to the sequential method, which peaks at 0.54. The PRNG only method again falls between the two, with a value of 0.45. These findings indicate that the proposed method effectively preserves the natural pixel correlations within the image, thereby reducing its susceptibility to sample pair-based detection techniques.

Similar performance trend can be observed in RS analysis and Fusion, where the advantage of the purpose method over the sequential method and PRNG only method is statistically significant. The performance improvement is slightly more apparent when the embedding rate is lower. At 25% embedding rate, the proposed method yields a 50% to 80% performance advantage compared to the PRNG only method for Sample Pairs, RS analysis, and Fusion; while at 100% embedding rate, the proposed method yields about 40% performance increase over the PRNG only method.

In addition to detectability, we evaluates the impact of data embedding on image quality using Peak Signal-to-Noise Ratio (PSNR). Higher PSNR values indicate less perceptible distortion, which is crucial for maintaining the integrity of the stego image. The proposed method using PRNG-based channel selection and adaptive matching technique consistently achieves the highest PSNR values across all embedding rates, beginning at 44.8 dB at 25% embedding and sustaining a competitive 38.8 dB even at full capacity. Both seqential method and PRNG only method exhibit similar PSNR trends, starting at 44.1 dB at 25% and declining to 38.1 dB at 100% embedding. The superior PSNR values of the prng proposed method indicate its enhanced capability to preserve visual quality, even under conditions of high payload embedding.

5 DISCUSSION

The sentiment expressed by Barradas et al. in their analysis of the Great Firewall (GFW)—"Far from being an impregnable fortress, the GFW's keyword censorship of HTTP is more like a Swiss cheese: overall solid, but full of holes"—aptly frames the broader significance of this work. It highlights the inherent vulnerabilities of censorship mechanisms and underscores the need for innovative, adaptable methods of bypassing such restrictions. The proposed PRNG-based

steganography algorithm contributes to this objective by offering a robust, high-capacity, and secure solution for transmitting sensitive information, even in heavily monitored environments.

5.1 Significance of Findings

5.1.1 Censorship Evasion. The results of the WeChat experiments demonstrate that steganography is a viable method for transmitting censored content over a known restricted platform. To the best of the authors' knowledge, this is the first study to demonstrate the use of image steganography to bypass censorship on WeChat, a widely used and heavily regulated network. The success of transmitting long-form text, including entire articles, without triggering keyword-based filtering, highlights the method's practical effectiveness.

The ability to bypass keyword filtering on a well-researched and actively censored network such as WeChat is particularly significant. It suggests that the proposed method can be generalized to other platforms employing similar content moderation strategies.

5.1.2 Steganalysis. The steganalysis experiments reveal the superiority of the proposed algorithm over baseline and intermediate methods. The incorporation of pseudorandom channel-based embedding and adaptive matching reduces detectability by approximately 50% compared to the intermediate approach. Moreover, the high PSNR achieved across varying payload sizes suggests that the method maintains perceptual transparency even under high-capacity conditions. This balance between security and image quality is critical for real-world applications, particularly when transmitting sensitive, long-form content like books or articles.

5.1.3 Practicality. A key strength of the proposed approach is its focus on user accessibility. By automating the decoding process and presenting a user-friendly interface, the web-based platform lowers the barrier to entry for non-technical users. The inclusion of Unicode character support further ensures that the tool can serve a global audience, enabling access to censored content in diverse languages and regions. This commitment to inclusivity is critical for addressing the global nature of internet censorship.

5.2 Conclusion

This work has introduced a novel steganographic algorithm and a practical web-based tool designed to circumvent internet censorship. By enabling the transmission of long-form content, such as books and essays, through steganographic images, we have demonstrated an effective and resilient method for bypassing keyword-based filtering on censored networks. The web application, with its lightweight design and decentralized deployment strategy, provides a framework for activists, bloggers, and community members to share censored literature while minimizing reliance on centralized infrastructure.

One of the key principles underlying this work is the importance of decentralization in combating censorship. Centralized platforms and larger entities often become high-profile targets for blocking efforts by censoring authorities. This makes it increasingly difficult to access content hosted on these platforms, as censors can apply sophisticated techniques such as IP-based blocking or traffic fingerprinting. In contrast, a decentralized network of independently managed sites avoids creating a single point of failure. Each site can operate autonomously, hosted by individuals or small organizations committed to sharing censored content with their communities. This approach not only enhances resilience but also aligns with the broader ethos of censorship resistance.

While the findings are promising, there are limitations to consider. First of all, the method has been tested on static image datasets and a single platform (WeChat). Additional testing is needed across different image types and platforms to validate its scalability. Secondly, the computational demands of client-side processing may pose challenges for users with lower-spec devices, particularly when handling large payloads.

Future efforts will focus on community outreach, recruiting activists to adopt and deploy this tool for localized content. An open-source repository will provide detailed documentation, preconfigured templates, and technical support to simplify setup and maintenance. Additionally, We are committed to working with volunteers to localize site templates, which ensures accessibility for diverse linguistic and cultural contexts, making the tool adaptable to global needs.

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